

Dexter Fernandes

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Robotics Engineer

Robotics Software Engineer specialising in perception, SLAM, and real-time deployment of computer vision systems on constrained edge hardware. Experienced in developing ROS2-based vision pipelines and deploying quantised CNN models across safety-critical and harsh field environments. Skilled in integrating LiDAR-camera data for localisation and mapping, with proven success delivering production-level AI modules for industrial and field-tested systems. Proficient in ROS2, OpenCV, PCL, Docker, and OpenVINO, with hands-on experience in Gazebo simulation, SLAM optimisation, and edge deployment for robust, autonomous operation under demanding real-world conditions.

WORK EXPERIENCE

Esbaa • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Developed and deployed production computer vision applications for real-time video analytics on edge devices delivering deterministic 24/7 operation in safety-critical oil-rig environments.
- Built an automated sensor-data cleansing pipeline handling 3.6 million records, reducing model rollout time from 3 days to under 24 hours.
- Implemented automated data ingestion, cleaning, training, and validation pipelines for large multimodal sensor datasets using MLflow, DVC, and Optuna, streamlining experimentation and reducing model iteration time by 40%.
- Developed real-time perception modules for oil-rig safety systems, deploying CNN-based vision pipelines with YOLOv5 on edge devices to improve detection accuracy by 15%.
- Refactored legacy monolith into Kafka-driven, Dockerised microservices, improving fault isolation and increasing MTBF by 35% across deployments.
- Designed and validated deterministic system behaviour through on-site field testing, performing root-cause analysis of model failures and reducing crash frequency by 67% in live video pipelines.
- Enhanced code quality and documentation standards by leading formal peer code reviews and quality-assurance activities within each sprint cycle, ensuring best practices.
- Owned end-to-end data pipelines, including automated data collection, augmentation, and labelling, increasing dataset diversity by 40% and reducing manual labelling effort by 60%.
- Documented deployment guardrails, failure modes, and recovery procedures, reducing cognitive load for engineers operating and maintaining fielded systems.
- Implemented continuous integration and testing workflows for edge inference modules, improving maintainability and deployment speed across hardware targets.
- Communicated experimental results and system performance to engineers and clients, translating quantitative metrics into actionable optimisation and deployment decisions.
- Integrated AI modules with production systems in collaboration with other teams to deliver fault-tolerant edge deployments.
- Engineered production-grade AI software for real-time perception and safety analytics, delivering deterministic performance across 24/7 oil rig operations.
- Validated deterministic model performance under extreme field conditions at a British Petroleum site, ensuring reliability across 24/7 operations.
- Applied Agile and Scrum methodologies to plan sprints, manage deliverables, and ensure iterative software improvement across multidisciplinary teams.
- Led on-site validation and diagnostics on oil-rig deployments, conducting root-cause analysis of model failures and improving overall system stability under environmental stressors.

- Built synthetic image generation and augmentation procedures to improve model robustness under harsh industrial conditions, reducing false positives and supporting long-term operational reliability.
- Presented key findings and metrics to leadership and clients to translate complex performance results into actionable insights for deployment and further optimisation.
- Built benchmarking and profiling harnesses to compare accuracy, latency, and stability pre- and post-optimisation, supporting measurement-driven iteration and regression detection.
- Diagnosed and resolved production failures through on-site analysis and structured validation checklists, reducing crash frequency by 67% in live video pipelines.
- Built a Django-based internal computer vision application with a web dashboard to visualise inference, system health, and performance metrics, enabling faster debugging.
- Designed and conducted adversarial-condition training experiments (occlusion, lighting variation, motion artefacts), benchmarking robustness via F1 and failure-rate metrics, achieving a 15% F1-score improvement within 3 months.

LivNSense Technologies Pvt. Ltd. • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI • Full-time

- Led technical delivery of real-time computer vision applications for object detection, pose estimation, and tracking, achieving sub-50 ms inference latency and reducing false positives by 33% on embedded hardware.
- Built and optimised deep learning pipelines with YOLOv5 and DeepSORT, enabling consistent object tracking and robust analytics in dynamic industrial environments.
- Deployed quantised vision models to rugged Intel VPU edge devices using OpenVINO, delivering deterministic real-time inference across RGB and IR sensor streams in harsh industrial environments.
- Re-architected the embedded inference pipeline and model optimisation workflow, increasing latency from 5 FPS to 15 FPS, enabling real-time detection of safety-critical events.
- Built automated labelling, dataset versioning, and performance-tracking pipelines, repurposing detection models for annotation and cutting manual labelling effort by 80%, accelerating iteration cycles.
- Developed synthetic data generation and augmentation tools to improve edge-case coverage and cut manual labelling time by 60%.
- Conducted systematic quantisation and optimisation experiments benchmarking accuracy–latency–memory trade-offs and reducing model size by 50% while meeting real-time constraints.
- Led a 4-engineer team to design, train, and deploy CNN-based object detection and tracking models for industrial safety applications, reducing false positives by 33% and achieving sub-50 ms inference latency on edge devices.
- Optimised neural network architectures using quantisation-aware training, structured pruning, and layer fusion reducing model size by 50%.
- Developed adversarial-condition training and validation conditions simulating occlusion, lighting variation, and motion artefacts, cutting performance variance by 25% and increasing resilience.
- Deployed a unified quantised CNN model for IR and RGB sensor streams using Intel OpenVINO, ensuring deterministic real-time inference on constrained edge devices.
- Collaborated with hardware, software, and QA teams to integrate AI modules into production systems, mentoring engineers on optimisation and deployment practices to improve delivery consistency.
- Engineered production-grade edge perception software for safety-critical environments, delivering deterministic real-time behaviour on Intel VPU devices.
- Mentored engineers and collaborated with hardware, software, and QA teams to integrate perception modules into deployed systems and improve delivery consistency.
- Presented key findings and metrics to leadership and clients, translating complex performance results into actionable insights for deployment and further optimisation.
- Built and maintained large-scale image and sensor datasets, automating labelling, versioning, and performance tracking to support anomaly investigation and continuous improvement cycles.
- Defined and standardised FP16/INT8 optimisation paths (including QAT, pruning, and layer fusion), reducing model size by 50% while preserving accuracy under tight memory and latency constraints.
- Implemented optical flow–based ego-motion estimation feeding motion representations into CNNs to classify vehicle state in production pipelines.

- Built asynchronous inference pipelines using OpenVINO to maximise throughput and determinism on Intel VPU edge hardware under real-time constraints.
- Evaluated edge versus cloud inference trade-offs to minimise bandwidth, latency, and operational cost while maintaining safety-critical performance, supporting production cloud-hosted ML workloads on Microsoft Azure.
- Designed data handling and retention practices aligned with GDPR principles, including data minimisation and controlled access, within production computer vision systems.
- Owned the full machine learning pipeline from data ingestion and labelling through training, optimisation, deployment, monitoring, and field validation, accelerating iteration cycles and improving deployment reliability.
- Provided technical leadership to a 4-member engineering team, setting experimentation standards, reviewing model performance, and guiding optimisation strategies for production deployments.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Radiological SLAM with LiDAR Odometry

- Designed and implemented a real-time SLAM system integrating LiDAR and radiological data for the teleoperation of the Clearpath Husky UGV development platform.
- Implemented ROS2 (rclcpp) point cloud pipeline in C++17 using PCL for filtering, segmentation, and registration.
- Developed and maintained ROS2 nodes and interfaces integrating perception outputs into a broader robotic system, managing sensor topics, tf2 frames, and message synchronisation for downstream localisation and navigation.
- Integrated perception pipelines into ROS-based robotic workflows, validating end-to-end data flow from sensors to SLAM and navigation components in Gazebo prior to deployment.
- Simulated and validated 2D/3D mapping and navigation workflows in Gazebo and ROS2 prior to real-world deployment.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

- Strong foundation in signals, systems, embedded hardware, and control, supporting later work in real-time AI and perception systems.

PROJECTS

ROS2 SLAM & Nav2 Simulation

Oct 2025

- Built a simulated mobile robot integrating 3D LiDAR and RGB camera data in Gazebo Harmonic to perform real-time SLAM and autonomous navigation using ROS 2 Jazzy.
- Applied slam_toolbox and Nav2 for 2D/3D mapping, localisation, and waypoint-based navigation with optimised Ceres backends.
- Processed multi-sensor data streams for spatial alignment and tested mapping accuracy across diverse simulated environments.
- Developed modular ROS 2 launch files and parameterised configurations for flexible experimentation with perception and navigation pipelines.
- Containerised the full application using Docker to ensure reproducibility, version control, and consistent deployment across development environments.

SKILLS

- **Perception & SLAM:** 3D Feature Extraction, Convolutional Neural Networks, Data Augmentation, Denoising, Eigen, LiDAR SLAM, NumPy, Object Detection & Tracking, Open3D, OpenCV, Point Cloud Library, ROS2 image_transport, SAM, Scan Matching, Scikit-Image, U-Net, Vision Transformer (ViT), YOLO

- **Cloud, Edge & MLOps:** Apache Kafka, AWS, Docker, DVC (Data Version Control), FastAPI, Intel VPUs, Microsoft Azure, MLFlow
- **Programming & Development:** Bash, C++17, CMake, colcon, Django, Gazebo (Harmonic), Git, Linux, Nav2, Pandas, PostgreSQL, Python, Qt, rclcpp, rclpy, ROS1 (Melodic/Noetic), ROS2 (Humble/Foxy/Jazzy)
- **Deep Learning and Optimisation:** Bayesian Optimisation, DeepSORT, Layer fusion, MultiRocket, ONNX, OpenVINO, Optuna, Post-training Quantisation, PyTorch, Quantisation-Aware Training, Structured pruning, TensorFlow, TensorRT
- **Edge AI and Inference:** Intel Core i7 (VTC7252-7C4IP), Intel Movidius Myriad VPU, NVIDIA Jetson AGX Xavier
- **Professional Skills:** Agile framework, Experiment-driven development, Stakeholder communication, Technical leadership